

HIMANSHU BANSAL

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OBJECTIVE:

Computer Science student with strong foundations in Deep Learning (**PyTorch**), Data Science libraries (**NumPy, Pandas, Matplotlib**). Seeking a remote AI/ML internship or entry-level opportunity to contribute to innovative AI-driven applications.

EDUCATION:

- Bachelor of Technology (B.Tech.) in Computer Science & Engineering
RIMT University, Mandi Gobindgarh, Punjab, India | 2023-2027
- Class XII | 2022-2023
- Class X | 2020-2021

SKILLS:

- Programming Languages: Python, C++, .NET (C#)
- Web Frameworks & APIs: FastAPI, Django
- Deep Learning & AI: PyTorch, Hugging Face, Scikit-learn
- Data Science and Visualization: Pandas, NumPy, Matplotlib
- LLM Frameworks: LangChain, LangGraph, FastMCP
- Developer Tools: Git, GitHub, Jupyter Notebook, Visual Studio
- AI Tools: Prompt Engineering, ChatGPT, Codex, GitHub Copilot

CERTIFICATIONS:

- Complete Machine Learning & Data Science | GeeksforGeeks
- Deep Learning for Natural Language Processing | Swayam
- Mastering Generative AI and ChatGPT | GeeksforGeeks

ACHIEVEMENTS:

- **Secured 1st place in three coding competitions** organized by the college (2026).
- Solved **50+ problems** on **LeetCode**, strengthening problem-solving and data structures fundamentals.

PROJECTS:

- **Multi-Agent Study Orchestrator**

Tech Stack: Python, FastAPI, LangGraph, LangChain, FastMCP, Gemma

- **Architected a LangGraph-based multi-agent system** orchestrating specialized AI agents for study note generation, concept explanation, and interactive quiz creation.

- **Integrated Model Context Protocol (MCP) tools** with LangChain agents and Gemini LLM to enable dynamic tool execution and stateful agent workflows.
- **Developed a FastAPI backend** with checkpoint-based conversation branching, real-time agent execution tracking, and asynchronous multi-agent communication.
- **Built a responsive web interface** supporting live workflow visualization, interactive quizzes, and personalized AI-assisted learning experiences.

- **Automated AI Response Judge**

Tech Stack: Python, PyTorch, FastAPI, Django, Hugging Face, DeBERTa-v3, Qwen 3.5

- **Architected a localized dual-pipeline evaluation system** separating subjective style analysis from factual correctness to reduce sycophancy bias in AI evaluation.
- **Engineered a preprocessing pipeline and fine-tuned a DeBERTa-v3 large encoder**, unifying heterogeneous datasets, enforcing a 512-token limit, and training with KL-divergence loss, label smoothing, and gradient accumulation.
- **Developed a Django–FastAPI microservice architecture** integrating Qwen 3.5-2B for zero-shot factual verification and asynchronous local PyTorch inference.
- **Outperformed the random baseline (33.3%)**, achieving **67.21%** accuracy on Anthropic HH-RLHF and **56.96%** on LMSYS Chatbot Arena evaluation splits.

- **Real-Time Face Recognition**

Tech Stack: Python, Django, OpenCV, NumPy

- **Built a Django biometric web application** to securely capture, register, and recognize multiple faces simultaneously in a live video stream.
- **Integrated deep learning models** using OpenCV's **FaceDetectorYN** (YuNet) and **FaceRecognizerSF** (SFace) for robust real-time face detection and embedding extraction.
- **Optimized the inference pipeline and engineered an offline embedding storage system** using JSON/Base64 APIs, NumPy (.npz), and JSON for fast, cloud-independent face matching.

- **Billing Management System**

Tech Stack: C#, .NET Framework, Windows Forms, SQL Server Express

- **Developed a desktop billing application** with customer management, inventory management, and GST-compliant invoice generation.
- **Implemented GST billing logic** supporting CGST, SGST, IGST, discounts, and automatic invoice calculations.
- **Integrated SQL Server Express** for persistent data storage and automated database initialization.